Word => söz, biz ögiydigan bolak, bosh joy bilan ajratiladi.

Token => modelning ögiydigan bolagi. U soz, sozning gismi, yoki bitta belgi bolishi mn

English text => 1 token ≈ 0.75 words.

Case 1: Token == Words.

Text=> "Cats sit on a mat"

words: cat, sit, on, a, mat => 5ta

tokens: cat, sit, on, a, mat => 5ta

Case 2: Token > Words

=> Uzun yoki kam uchraydigan söz bolaklarga bolinadi.

=> Tinish belgilari alohida token bolaoli

=> ozbek, koreys kabi tillar koprog tokenga b-di.

Text: "Deconstitutionalization"

words => 1ta

tokens => De, consti, tutional, iza, tion. 5ta

Case 3: Token < words.

Text: "United States"

words: United, States => 2ta

tokens: United States => 1ta.

- bu sozni bir biridan ajratsa maiko golmaydi.

lugatda bu bitta butun, söz.

=> English - 1 token ≈ 0.75 words.

=> Spanish / French - 1 token ≈ 0.50 words.

=> Russian - 1 token ≈ 0.28 words.

=> Japanese / korean - 1 token ≈ 0.18 words.

Ingliz tili uchun token / so'z kam, Chunki tokenizer shu tilga o'qitilgan.

LLM model training qilishdagi oldin hisoblash kerak bo'lgan lar.

(1) Parameters:

$$12 \times L \times d^2$$

- Layers => L -> transformer layerlar soni.
- d => hidden dimension, ya'ni bitta token'ni ifodalovchi vektor o'lchami.
- d=768 => bitta token model ichida 768 ta sonli qiymatdan iborat vektor bilan ifodalanadi.
- bitta layer uchun taxminan $12d^2$ parameter chiqadi.

Masalan: "Hello world" => ["Hello", "world"].

Agar d=4 bo'lsa.

Model "Hello" tokenini quyidagicha ifodalashi mumkin.

Token: Hello			
0.2	0.7	-0.1	0.5

4 ta son -> d=4.

Agar: d=768 bo'lsa 1 ta token 768 ta son bilan ifodalanadi.

12 => attention + FFN matritsalar hisobi.

Nega d^2?

Model shu vektorni o'zgartirish uchun (weight matrix) ishlatadi. $d=4$ bo'lsa matrix o'lchami $4 \times 4 = 16$. ta weight bo'lishi mumkin. $d^2 = 16$

Jami parameters

$L = 24$ $\quad 12 \times d^2 \times L$

$d = 2048$ $\quad 12 \times 2048^2 \times 24 \approx$ 1.21 milliard.

② Xotira sig'imi:

16 B × N.

Weights (FP32) – 4 bayt
Gradients (FP32) – 4 bayt
Optimizer states – 4 bayt
Activations – 4 bayt
} 16 bayt

Memory = N × har parameter uchun bayt

Memory = 16 × 1.21 milliard ≈ 19.3 GB.

Agar bizda 320 GB mavjud bo'lsa, taxminan 6%

③ Compute budget

$C = 6 \cdot N \cdot D$

- N – model parameters soni.
- D – training tokenlar soni.

Chinchilla qoidasi bo'yicha $D \approx 20 \times N$

- Forward = 2, Backward = 4. → 6ND.

$C = 6 \times 1.21$ milliard $\times$ 24.2 milliard

1.75×10^{20} FLOPs

③ Vaqt (wall-clock)

$$t = \frac{6ND}{GPU \cdot peak \cdot MFU}$$

- peak - A100 BF16 = 312 TFLOPs/s
- MFU - real samaradorlik, A100 da 40%.
- GPU - 4 ta parallel ishlay.

$t = 1.75 \text{ kro} / (4 \cdot 0.40 \cdot 312 e12) \approx 4 \text{ kun}$.

4 kun ($\approx$ 97 soat).

④ Cost.

$$Cost = GPU_{soni} \times soat \times GPU_{narx/soat}$$

GPU soni - 4 ta,

GPU narxi - $1.5-2 / soat

Cost: $97 \times 4 \times 1.5-2 \approx \$600-800$

Bitta toliq o'qitish sikliniq narxi.

Trades off => biron narsaga erishish uchun biron narsadan voz kechish.

Parameterlar	$N \approx 12 \times L \times d^2$
Xotira (Adam W)	16 bayt $\times N$
Compute	$C = 6 \times N \times D$
tokenlar (Chinchilla)	$D \approx 20 \times N$
Vaqt	$6ND / (GPU \cdot peak \cdot MFU)$
Narx	soat $\times$ GPU $\times$ $/soat

Params => Xotira => Compute => vaqt => Narx => samaradorlik

Trainingdan oldin.

Sonlarni kompyuterda ifodalash uchun ishlatiladigan (data formatlar). Ular sonning saqlash xotira egallashi va hisoblash aniqligiga ta'sir qiladi.

⇓

FP32 => 32 bit Floating point.
FP -> Floating point, ya'ni kasrli sonlarni ifodalaydi.
Xotirada har bir soni 4 Byte.

1e-38 ... 1e38.

FP16 => 16 bit => 2 byte -> 6e-5 ... 65504.
BF16 => 16 bit => 2 byte -> FP32 bilan birxil

Bitlar taqsimoti:
FP32 -> 32 Bit
Sign -> 1 bit, Exponent -> 8 bit, Fraction - 23 bit.
FP16 -> 16 bit.
Sign -> 1 bit, Exponent -> 5 bit, Fraction - 10 bit.
BF16 -> 16 bit
Sign 1 bit, Exponent -> 8 bit, Fraction - 7 bit.
Sign -> Son musbat yoki manfiyligini aniqlaydi.
Exponent -> Sonning kattalik diapazonini aniqlaydi.
Fraction -> Sonning aniqligini aniqlaydi.

=> Trainingda gradientlar juda kichik bo'lishi mumkin. (10^{-8}). FP16 da bu aniq xolga aylanadi. Chunki ~~Fraction~~ diapazoni kichik

$6.1 \times 10^{-5} > 10^{-8}$ => gradient -> 0

BF16/FP32 da to'g'ri saqlanadi. diapazon katta

Xulosa: BF16 => tezlik, FP32 = xavfsizlik
FP16 -> xavfli

Param

Model weights (FP32) -> 4 byte
Gradients (FP32) -> 4 byte.
Adam m (FP32) -> 4 byte.
Adam v (FP32) -> 4 byte.
Jami (Adam W) -> 16 byte.